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Design the airline's disruption assistant

The system. A storm closes the airline's hub for the evening. Every passenger on a cancelled flight opens the app and asks what they can do. The assistant answers within 5 seconds from the airline's own documents, as things stand right now, quoting the rule or waiver it relied on.

SOURCE	CHANGES	WHAT IT IS
fare rules	monthly	2,000 documents
waivers	hourly	routes, dates, what the passenger may do, until when
flight status	minute	1,500 flights: status, gate, rebooking options
booking	per request	this passenger's flight, fare class, what they have been offered

Task.

1. **Architecture.**
   - Draw the whole system; show how each source reaches an answer, and how old it can be
2. **Measurement.**
   - Per stage, before any passenger sees it: what is measured, against what
   - The unanswerable third: where those questions come from; what the floor stops, what the model answers anyway
   - What decides a release, and one guardrail that can stop a release whose primary metric rose

Your design should handle:

1. Each source at its rate: indexed, summarized ahead, or looked up at the request
2. A waiver as a document: its chunk, its metadata, what happens the minute it expires
3. Ten thousand passengers asking the same question in an hour: the cache and its key
4. Money is quoted verbatim with the waiver's id: what checks that, on every answer

(Bonus) At the storm's peak 150 questions a second arrive, and the language-model service can serve 50 a second: what does the passenger still get, and what does the system never do?

One question. At 19:40 the assistant promised a hotel under a waiver withdrawn at 19:30. What does the log hold, which component let it through, which should have stopped it?